



Evolution Paths for Knowledge Discovery and Data Mining Process Models

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Abstract

Despite the hype around data analytics, the success rate of analytics initiatives remains very low and the value of data in organisations is left hidden. Various research studies show that the main barriers to analytics adoption are organisational and the lack of structured approaches on how to conduct analytics initiatives is a possible cause of analytics project failures. Data mining process models then become fundamental means to support analytics project management and minimise the risk of data dredging. In this paper, Knowledge Discovery and Data Mining process models are reviewed starting from the most popular models currently in use. Four distinctive research paths for data mining process models have emerged. These evolution paths seem to address limitations of the CRISP–DM model which remains the *de facto* standard in industry. The research streams identified include the evolution of the human role; the relevance of iteration and interactions; the role of data and knowledge repositories; and the integration of software engineering/agile methodologies. In the future, these four research streams should be combined to support the development of more encompassing process models.

Keywords Data analytics · Knowledge discovery in databases · Data mining · Process models · CRISP–DM

Introduction

The value of Data Analytics (DA) is widely recognised by the industrial and academic community as real and significant. Big data is revolutionizing all aspects of our lives ranging from enterprises to consumers, from science to government [1]. This transformation is occurring at every level, from the micro-economic to the macroeconomic, from the way individual stores stock their shelves, to the way manufacturers and their customers connect to realise new forms of value, to the way economies and societies function [2]. Symptomatic of the ever-growing popularity of big data and data analytics is the attention paid to these subjects not only by the scientific and industrial literature but also by many news media [3, 4].

There is a widespread belief in companies' management that analytics offers value and represents a fundamental source of competitive advantage within their own industry. For those companies that lack available analytics insights in

their systems, the awareness that the potential hidden behind available data is not fully exploited generates thrust to the adoption of advanced information approaches [5]. This craving for analytics is incentivized by the many success stories available in the literature. However, turning a world full of data into a data-driven world is an idea that many companies have found difficult to pull off in practice [6]. Despite the ever-growing interest that organisations show towards data science and its recognised value on business performance, advances in the adoption of big data and analytics in organisations of different sizes and sectors are still very slow [7]. Although one clear reason cannot be found, and a different set of issues are experienced by organisations depending on their geography and current level of maturity, a survey conducted by Gartner in 2018 [7] identifies three most common barriers to analytics adoption:

- Determine how to get value from projects;
- Defining data and analytics strategy;
- Solving risk and governance issues.

The first barrier in the list above is mostly related to management issues and is a recurrent and fundamental barrier reported by surveys conducted in the last few years. For

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instance, LaValle et al. [5] also reported that “*The leading obstacle to widespread analytics adoption is lack of understanding of how to use analytics to improve the business*”; the same survey suggests that the adoption barriers that organizations face most are managerial and cultural rather than related to data and technology. This has not changed in time as suggested by more recent surveys [6, 8, 9], including Dykes [10] that identifies technology as the last roadblock to successfully becoming a data-driven business; other roadblocks include leadership, resources, process and culture.

Scepticism around the real value of analytics is also deterring management from investing in analytics initiatives. This clearly emerges in a survey conducted internally among partner companies of the Irish Centre for Manufacturing Research (ICMR); the survey results suggest that, although all companies rate data analytics as of key importance for their future, data analytics does not score as well when it comes to addressing what technology among those enabling the fourth industrial revolution will help fix the problems that the companies are faced with today (Fig. 1). This is probably due to the difficulty in quantifying return on investment (ROI) of analytics projects which in turn leads to difficulties in finding executive sponsorship; often the perceived costs of analytics initiatives outweigh foreseen project benefits [5, 6].

Another managerial barrier is the lack of understanding around where to start and how to manage an analytics project, which also translates into a lack of understanding on how to go about solving problems using data science [11, 12]. While the focal attention of companies is currently around big data—how to collect it, how to develop data infrastructures, how to store and access data—their lack of focus around creating value out of data, analysing that data and designing analytics strategies around it is leading to poor performance metrics with respect to data projects. In fact, a mere 13% of data and analytics projects reach completion, and of those that do, only 8% of company leadership report being completely satisfied with the outcome [13].

Following a clear process model and methodology while addressing a data problem becomes fundamental to minimise the risks associated with “data dredging”. A blind application of data science techniques can lead to discovering meaningless patterns, and this can be prevented from occurring by formally structuring Knowledge Discovery and Data Mining (KDDM) processes [14]. Although it is likely that typical process steps, such as business objectives identification, problem definition, stakeholder identification, data architecture selection, analytical modelling and results validation, will naturally emerge through a data analytics project, the lack of a well-defined process model makes projects error-prone and devoid of any structured learning mechanism for future initiatives [15]. KDDM process models guide data project stakeholders to stay focused on the end product and prioritize goals for data analysis.

The value of a formal process model does not collide with the open-ended nature of a data project (i.e. try to find “value” in the data) as a robust methodology will support effective project planning and achieving project management goals such as on-time project delivery, and group coordination/ communication [16]. The academic focus is more on developing standards in terms of languages to use in data mining and analytics, whereas the industry focus is on defining methodologies that can guide the implementation of KD applications [17].

A key challenge for twenty-first-century data analytics is the development and dissemination of widely accepted standards, including process models, that, if adopted, will stimulate major industry growth and interest [18, 19] and will provide solid base for future research [20]. In order to support the definition of comprehensive process models, existing models are reviewed in this paper and research patterns that have guided the development of models in more recent years are identified so that directions for future research can be drawn.

In the remainder of this paper, the terminology related to knowledge discovery and associated tools is illustrated

Fig. 1 Enabling technologies versus opportunities

		Enabling Technologies										
	Technology Score	17	16	29	21	37	23	20	3	21	16	22
Value Score	Opportunity Category	Cyber Sec	Cloud Comp.	IOT	Data Analytics	Adv. Automation	Adv. Sensorisation	Mobile Connectivity	Virtual & Augmented Reality	Cobotics	Additive Manufacturing	Energy Management & Storage
6	Asset Utilisation ³	0	4	12	2	19	6	7	0	15	1	9
7	Labour ³	0	0	5	10	17	4	0	0	14	0	0
12	Quality ²	0	0	4	4	21	13	0	4	16	5	2
6	Supply/Demand Matching ¹	4	4	11	0	3	0	0	0	4	0	0
5	Resource/Process ³	4	0	12	10	10	0	0	0	8	0	10
8	Time to Market ¹	0	0	3	0	7	0	0	0	5	7	0
10	Service/After-sales ¹	0	0	5	0	0	0	5	0	0	0	0
6	Traceability ^{1,2}	0	0	7	2	0	0	0	0	0	0	0
14	Flexibility	4	4	5	0	10	4	4	0	15	5	6
13	Efficiency	0	0	9	2	24	2	0	4	16	5	18
		12	12	73	30	111	29	16	8	93	23	45
	Super Category											
11	1 - Supply Chain	4	4	4	4	5	3	6	4	7	5	0
7	2 - Regulation	0	0	0	0	3	5	0	0	0	5	3
15	3 - Scalability	4	5	5	0	5	3	9	0	8	5	3

in “[Terminology](#)” section. Relevant frameworks/process models available in the academic and industrial literature are reviewed then. Classic models are introduced in “[Classic KDD Process Models](#)” section, limitations to the most popular model are identified, and process models developed to address those limitations are discussed in “[Data Mining Process Models Evolution](#)” section. A comparative analysis of process models is presented in “[Comparative Analysis](#)” sections, and concluding remarks are drawn in “[Discussion and Conclusions](#)” sections.

Terminology

Before analysing currently available process models in knowledge discovery, it is opportune to clarify their meaning and origins. Frameworks, process models, methodologies, techniques and tools are essential strategic, tactical and technical elements for developing a data analytics project. Definitions of these elements are provided in this section.

Knowledge discovery in databases (KDD)

This is *the nontrivial process of identifying valid, novel, potentially useful and ultimately understandable patterns in data* [21]. Being a process, KDD is comprised of many steps, which involve data preparation, search for patterns, knowledge evaluation and refinement, all repeated in multiple iterations [21]. Data mining represents a mere step in this process and is concerned with the algorithmic means by which patterns are extracted from data [21]. The term KDD was introduced in 1989 to highlight that the end product of data-driven discovery is knowledge. Currently, the term Knowledge Discovery and Data Mining (KDDM) is used to refer to the knowledge discovery (KD) process applied to any source of data. KDDM concerns the entire knowledge extraction process, including how the data is stored and accessed, how to develop efficient and scalable algorithms that can be used to analyse massive data sets, how to interpret and visualise the results and how to model and support the interaction between human and machine [14]. The DIKW (Data Information Knowledge Wisdom) pyramid (Fig. 2) suggested by Jifa and Lingling [22] schematizes the process of knowledge discovery. Each layer in the hierarchy adds certain attributes over and above the previous ones. Data is the most basic level; it is raw and does not have meaning by itself; information adds context (i.e. it gives meaning to data by adding relational connections, but it is not necessarily useful); knowledge adds how to use information (i.e. collects useful information); wisdom adds when and why to use it (i.e. it is the result of understanding knowledge and extrapolates new knowledge from previously held knowledge).



Fig. 2 DIKW pyramid

Framework

Framework is defined as a “a basic structure underlying a system, concept, or text” (“Oxford Dictionary” 2019). Its definition is a little loose, and this reflects its purpose that is to provide structure and direction on a preferred way to realise defined results/goals without being too detailed or rigid. In essence, a framework provides guidelines while maintaining a high degree of flexibility so that it can be adapted to different conditions. Its flexibility also concerns the choice of using part of it or its entire content while conducting a project.

Process model

A process model is the set of framework activities and tasks to achieve results/ goals within a project. A process model also specifies inputs and outputs of each task. The final objective of a process model is to make a process manageable, repeatable and measurable; characteristics of a good process model are [19]:

- Effective—a process model should help achieve intended objectives;
- Maintainable—issues within the process model should be easy to identify and correct;
- Predictable—the process model should provide support for effective project planning;
- Repeatable—the process should be sufficiently generic so that it can be applied several times in different contexts;
- Improvable—a good process model should be amenable for improvement in different aspects;
- Traceable—a process model should offer support to follow the status of a project.

Methodology

Methodology is defined as *a documented approach for performing activities in a coherent, consistent, accountable and repeatable manner* (TEAF, 2000). It comprises a defined set of rules, methods, test activities, deliverables and processes that serve to solve a specific problem. A methodology

defines not only tasks, inputs and outputs but also the way in which tasks should be carried out. A methodology can be an instance of different process models and vice versa. For each task, techniques are chosen which define how to execute the task. Tools may be used to apply techniques and make the tasks easier. In summary, a process model clarifies what to do, whereas a methodology defines how to do [19].

Life cycle

Life cycle determines the order in which tasks will be performed and establishes the way to develop a project. It defines the order of phases and processes and identifies the criteria that will be used to progress from a phase to the next.

Classic KDD process models

Considering that process models generally provide insights into the dynamics of a project and detail the steps involved to achieve success more so than frameworks, research has focused on the development of process models. Some of the most popular models are illustrated in this section; their evolution over time is discussed in terms of main research streams that have stemmed from limitations identified in the seminal models. It is worth noting that although the models discussed in this section are typically classified as process models, some of them are generally referred to as frameworks (see, for instance, the CRISP-DM model in “CRISP-DM” section). This shows the practical equivalence of terms such as framework and process model within a business and industrial data analytics context.

KDD model

The foundation of KDD process models was laid down with a publication by Fayyad et al. [21]. The KDD model they developed did not refer to any specific data mining (DM) technique but provided support for the complex and iterative process of knowledge generation [14]. The KDD model comprises five steps from a data viewpoint (Fig. 3): data

selection and sampling, data processing, data transformation, data mining and evaluation. In the data selection step, the analyst creates a target data set from a large data store; the selected data is relevant to the knowledge discovery process. Data processing consists of dealing with noisy and missing data; it ensures that correct input is used in the KDD process so that a valid output is obtained. In the data transformation step, dimension reduction and transformation methods are used to identify useful attributes. The data mining step involves choosing the data mining task that matches the analysis goals, choosing algorithm(s) and method(s) with corresponding parameters and applying these methods to extract patterns from data. Finally, the evaluation step aims at interpreting the mined patterns and extracting useful knowledge from them; corrective actions with subsequent iterations may follow.

The KDD model is fundamentally a data-centric model that emphasises the iterative and interactive nature of data analysis tasks. However, while data-related tasks are detailed, there is a lack of business perspective in this outline model [23]. In the detailed version of the model, a pre-processing and post-processing stage is added. The initial stage focuses on developing an understanding of the application domain and identifying the project goals from a customer’s perspective; the final stage concerns the consolidation of the discovered knowledge, which should be integrated in appropriate systems and documented.

Nevertheless, the lack of business focus in the KDD model remains evident considering that there is no deployment phase through which actual validation of data mining results would be carried out. Despite this, the relevance of the KDD model as a reference model for analytics project is apparent as suggested by a survey conducted in 2014 [24] that shows that the use of the KDD process in analytics has remained stable in the last decade (Fig. 4).

This survey also reveals that other two classic KDDM models, namely SEMMA and CRISP-DM, which will be discussed next, remain popular. Most interestingly, no companies reported using no methodology which is a clear indication that “data dredging” is no longer a practice in the era of big data and there is always an attempt to follow

Fig. 3 KDD process model [21]

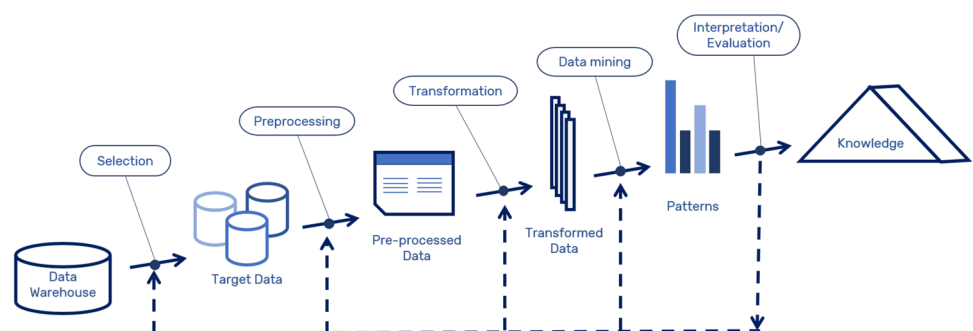


Fig. 4 Process models used for analytics/DM/data science projects (200 votes in total) [24]

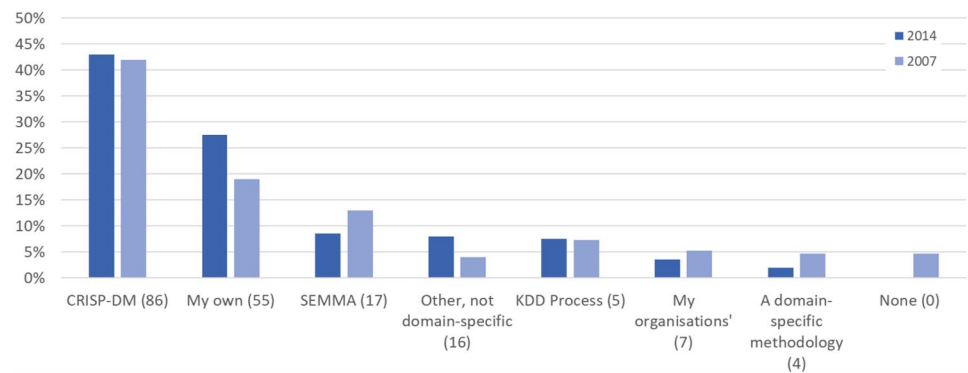


Fig. 5 SEMMA process steps



a process model by companies that aim at conducting analytics projects effectively and efficiently. Likewise, the fact that the use of domain-specific methodologies is decreasing is a reflection of the need of standardising KD processes [17]. The surge in the use of other not domain-specific methodologies and own methodologies is probably an indicator of the maturity of data analytics initiatives and the result of experimentation with different process models by both data analysts and business management; people involved in analytics projects tend to experiment with available methodologies and select steps from different models based on their own needs.

SEMMA

The SEMMA process model was introduced by SAS in 1997; it comprises five steps: sample, explore, modify, model and assess (Fig. 5), which are focused on the model development aspects of data mining [19]. It can be viewed as an implementation of the KDD process model as there is a perfect parallel with the tasks of the KDD outline model. The business understanding phase is clearly missing; however, as argued by Azevedo and Santos [17], it could be included in the sample phase since the data cannot be sampled until there is a clear understanding of both the application domain and the customers' goals. Likewise, the deployment step is not explicitly defined, but it represents the main reason to embark on a data analytics initiative. The SEMMA steps are directly linked to the SAS Enterprise Miner software, and this probably justifies the presence of SEMMA among the top process models used for data analytics initiatives (Fig. 4)

Although the SEMMA model can be iteratively used, there is no clear evidence of the presence of iterations and interactions between tasks in the original model. These highlighted shortfalls of the model and probably its decreasing popularity [24] have brought SAS to revise their process

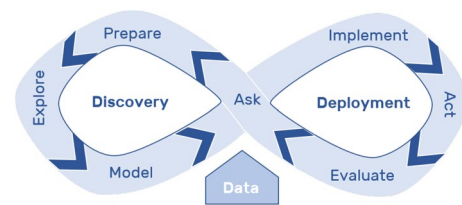


Fig. 6 SAS analytical life cycle [25]

model and suggest a new version of it in 2016. The resulting analytical lifecycle model (Fig. 6) is presented as an iterative process that can be used to produce repeatable and reliable predictive results [25]. Two phases are highlighted in this model—discovery and deployment. The discovery phase is mainly based on the SEMMA steps and includes:

- *Ask a question*—equivalent to business understanding and business goals definition;
- *Prepare the data*—it focuses on joining data from different sources and transforming it into valid input to analytical models; this step is still the most time-consuming (i.e. 80% of the discovery phase);
- *Explore the data*—initial data analyses using simple visualisation tools can help refine the business questions, select applicable analytics approaches and identify missing data elements;
- *Model the data*—numerous analytical and machine learning modelling algorithms are applied to the data to find the best representation of the data relationships that will help answer the business questions. Model “tournaments” (i.e. concurrently create hundreds or thousands of models and run tests on them) can be used to automate the process of selecting the best modelling approach.

The deployment phase includes:

- *Implement your models*—insights learnt from the discovery phase are put into action using repeatable, automated processes. This is a critical step where an analytical project generally slows down due to the transition of the model's ownership from data scientists to IT architects. Testing, validation and approval of the model are part of this step;
- *Act on new information*—Once the model is approved, strategic and operational decisions can be made. Strategic decisions are usually taken by humans based on analytics results; operational decisions can be automated and coded into the production system so as to be objective, consistent, repeatable and measurable;
- *Evaluate your results*—the outcome of the actions suggested by the analytical model is evaluated, and the feedback is used for the next generation of the model;
- *Ask again*—this step closes the loop of the iterative process model and is based on the consideration that even accurate models are not forever and require updates due to dynamic environmental conditions.

This model is still based on classic KDD process steps but emphasises the dynamic and iterative nature of an analytics project and highlights the interaction between the discovery and deployment phase. Iterations and interactions are deemed key elements of effective analytics frameworks and process models [26] as discussed in “[Iterations and Interactions](#)” section.

CRISP-DM

The CRISP-DM (CRoss Industry Standard Process for DM) was initially proposed in 1996 by a consortium of four companies: SPSS (a provider of commercial DM solutions), NCR (a database provider), Daimler Chrysler and OHRA (an insurance company). The last two companies served as sources of data and case studies [14]. Its first version was officially released in 2000 [27]. CRISP-DM is a hierarchical process model (Fig. 7) based on a waterfall life cycle and has a methodological component, as it gives recommendations on how to conduct some tasks.

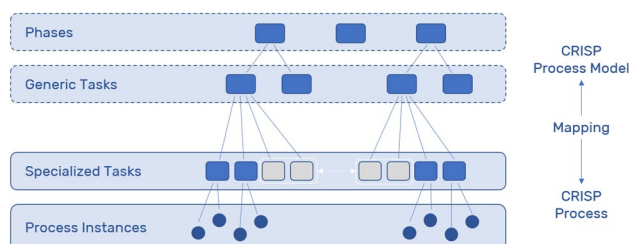


Fig. 7 CRISP-DM hierarchy [27]

In the top level of the CRISP-DM hierarchy, there are six phases (Fig. 8): business understanding, data understanding, data preparation, modelling, evaluation and deployment. Each phase consists of generic tasks for which inputs and outputs are clearly identified; the generic tasks are complete, in the sense that they cover the whole DM process and all DM applications, and stable, in the sense that the model would still be valid following unforeseen technology developments. Although the lifecycle model is predominantly linear and sequential, iterations and feedback loops are considered; the arrows in Fig. 8 indicate the most important and frequent dependencies between the phases. Below the generic tasks, specialised tasks (i.e. actions taken for the generic tasks) and process instances (i.e. actions, decisions and results obtained for specific DM initiatives) complete the hierarchy (Fig. 7).

CRISP-DM is possibly the best documented process model, and its detailed documentation has probably played a fundamental role in making it become the *de facto* industry standard in terms of KD process models (see Fig. 4). CRISP-DM takes the most interesting elements of previous KD models, and it is a recurrent reference for all the models that followed it. CRISP-DM explicitly introduced two fundamental steps: business understanding and data understanding [23], which are considered the cornerstone of any successful data mining project as they can help dig more insights into the business objectives and current data availability [28]. In view of the research path evolution of process models, main limitations of the CRISP-DM model emerge:

- the lack of consideration of the involvement of human resources in the analytics process;
- the predominantly sequential nature of its life cycle which concerns both the lack of clearly defined iterations and interactions between tasks, and the lack of a continuous improvement phase that follows the model

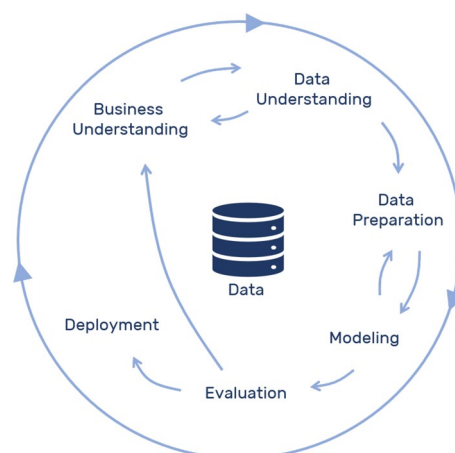


Fig. 8 Phases of the CRISP-DM reference model [27]

deployment and closes the feedback loop with business processes;

- the lack of consideration of alternative data sources;
- the lack of integration of software development models.

In response to these issues and evolving business needs, a CRISP-DM 2.0 special interest group (SIG) was established in 2008; the group extended the original consortium to include vendors, service providers, researchers, end-users; however, no updated model has been released yet, and the CRISP-DM website has also been shut down. Other authors have addressed the issues identified above, and various models inspired by CRISP-DM have been proposed in the literature. A selection of these relevant models is presented in the next section.

Data Mining Process Models Evolution

This section elaborates on the evolution of KD process models based on the CRISP-DM limitations identified above. More specifically, “[Relevance of Human Role](#)” section considers how the relevance of the human role in analytics process models has evolved through the years; “[Iterations and Interactions](#)” section highlights the importance of interactions and iterations in effective process models and illustrates some relevant models characterized by iterative loops and clearly defined tasks interactions. “[Data-Focused Process Models](#)” section reviews models that introduce data structure and knowledge repository concepts to support effective knowledge management. Finally, “[Towards Agile Models](#)” section introduces models that are based on the integration of project management tasks that are typical of software engineering with analytics process steps.

Relevance of Human Role

The relevance of the human role in a KD process is certainly not a recent topic in KD and has been addressed since when the KDD concept was first introduced by Fayyad et al. [21]. Indeed, in parallel with the data-centric KDD model, Brachman and Anand [29] suggested a human-centric process model that emphasised the interactive involvement of a data analyst during the process. The human-centric process was defined as a series of knowledge-intensive tasks consisting of complex interactions, protracted over time between a human and a large database possibly supported by a heterogeneous suite of tools. The importance of involving other key actors beyond the mere data analyst in the KD process was first suggested by Lee and Kerschberg [30] that introduced the “critique by a panel of experts” as the final step of the KD process in their Knowledge Discovery Life Cycle model (Fig. 9); the critical feedback of domain experts

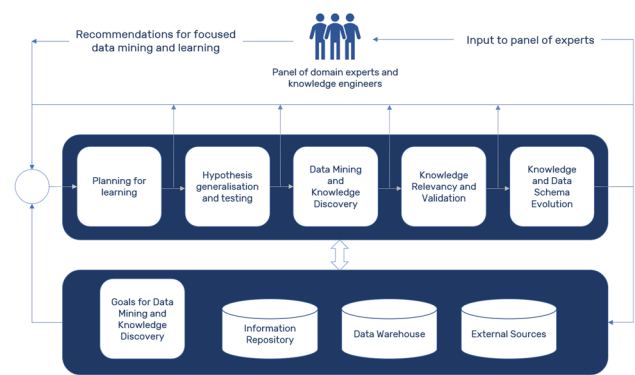


Fig. 9 Knowledge Discovery Life Cycle (KDLC) process model [30]

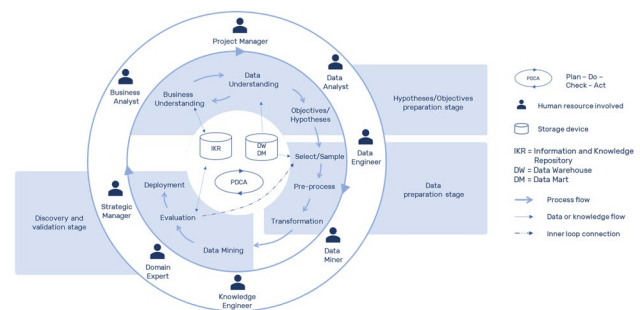


Fig. 10 Data Mining Life Cycle—DMLC [28]

would trigger the refinement of the project scope and identify more focussed DM objectives. Human resource identification was highlighted as a fundamental pre-processing step in the model proposed by Anand and Büchner [31]. Another human-centric model was proposed by Gupta et al. [32]; it was based on a data-centric model with extensions to provide support for experimentation and monitoring activities [14].

In 2002, Hofmann modified the CRISP-DM model to address some of its weaknesses [28]. In the Data Mining Life Cycle (DMLC) model suggested by Hofmann and Tierney [28] (Fig. 10), the role of various types of human resources involved in each process step is identified; various data sources, such as data warehouses and data marts, are introduced; a knowledge repository where all learnings from the process can be collected and used to create and update business strategies is mentioned and the whole model is based on the Deming’s cycle for continuous improvement. The attempt to simplify the KD process is also evident as Hofmann and Tierney [28] identify three main stages: hypothesis/objectives preparation, data preparation and discovery and validation (Fig. 10); this summarising approach is also used by Rogalewicz and Sika [33] that suggest a three-stage model that encompasses other available models (Fig. 11). The main limitation

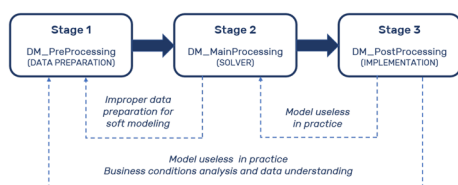


Fig. 11 Diagram of approach to chronological knowledge discovery process [33]

of the DMLC model is related to the fact that it mainly focuses on large-scale data projects for which various roles are required and it is not suitable for small analytics initiatives.

Also focused on large data analytics projects is the RAMSYS model [34] that adapts the CRISP-DM model to projects where interested parties are geographically dispersed but work together on the same problem in a collaborative way (see “Iterations and Interactions” section for more details on this model). In most KD process models, the domain expert role is generally limited to supervising, consulting and reviewing the work done by a data analyst [35]. Clear evidence that most process models are created from a data analyst perspective is provided by the nature of the first two typical steps (i.e. understanding business and understanding data); if the domain expert was actively and constantly involved in a data analytics project, these steps would probably be redundant. However, in knowledge-intensive fields such as the biomedical realm, significant domain expertise is required to conduct data mining and interpreting results, from a safety viewpoint of (medical) knowledge discovery and also considering that human knowledge includes information that domain data may not have. Hence, the integration of domain experts more deeply in data mining and machine learning tasks is critical. Knowledge elicitation from domain experts and machine learning should be seen as distinct approaches for knowledge discovery with different and complementary capabilities. Means to facilitate the interaction of non-IT experts with data analytics software should also be catered for. In the model proposed by Girardi et al. [35], which is an adaptation of existing models, the core element is the switched role of the domain expert from a marginal role to a central role. In this model, there is no strict sequential order for the steps and tasks can be performed in parallel with feedback loops activating corrective actions on previous steps (Fig. 12). For instance, data validation and data analysis may cause re-acquisition or revision of existing data. In order to facilitate the integration of domain experts in data mining activities, Girardi et al. [35] suggested recourse to domain ontology as fundamental data structures in the data modelling phase; moreover, they also develop a very high degree infrastructure that automates data pre-processing and algorithm parameterization to allow

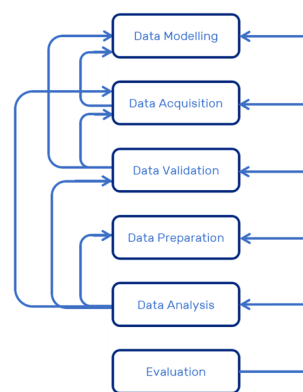


Fig. 12 A process model for domain expert-centred knowledge discovery [35]

a non-IT user to apply DM algorithms by leveraging the ontology-based meta-information.

Iterations and Interactions

Another recurrent topic in KD process models is the importance of iterations and interactions. Iterations concern the repetitions of tasks within the process model using feedback loops to correct or refine assumptions resulting from the application of previous steps; they also refer to the repetitive application of the entire model to activate continuous improvement and update the model based on evolving business conditions.

Although most models include iterations between various phases, some models pay particular attention to this aspect. For instance, Collier et al. [36] were the first ones to introduce the re-iteration concept by transforming the KDD model into a continuous cycle (Fig. 13); the deployment or re-iteration phase works as an inspection phase to the entire project and suggests that whenever the project objectives are not reached, a new iteration of the entire cycle should be triggered; however, this model lacks an internal loop that

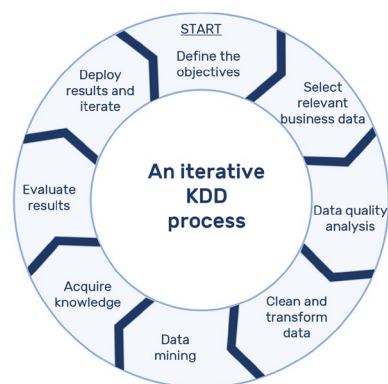


Fig. 13 Refined KDD process model [36]

would allow for corrective actions before the deployment phase. The DMLC model by Hofmann and Tierney [28] (Fig. 10) is also based on the PDCA (Plan, Do, Control, Act) cycle and introduces internal iterations based on the outcome of the evaluation step. The DMIE (Data Mining process for Industrial Engineering) model proposed by Solarte [37] explicitly introduces for the first time an “on-going” support phase, which focuses on support and maintenance activities that involve data backups, data maintenance, data mining model updates and software updates (Fig. 14).

In the Data Science Workflow model introduced by Guo [38], four stages are identified: preparation, analysis, reflection and dissemination (Fig. 15). The analysis phase is purely based on iteration cycles. There are also iterations between the analysis phase (based on programming) and reflection phase (based on thinking and communication) as much of the analysis process is recognised to be a trial-and-error process. Practical insights are given by Guo [38] that highlights that, given a fixed project duration, the speed of each analysis cycle determines the amount of information that can be gained by time unit; slow-down factors to these processes include absolute running times

(large amount of data or slow algorithms), incremental running times (small changes that affect only part of the code), crashes from errors. The main limitation of this model is that evaluation (explicitly defined) and business understanding are missing.

The CRISP-DM model has been revised by Rollins [11], a scientist working for IBM, that developed the Foundational Methodology for Data Science (Fig. 16). The flow of this methodology illustrates the iterative nature of the problem-solving process; the feedback control, introduced as the last step, is used to make sure that the model, once developed and deployed, is not abandoned to itself, but it adapts to changes and conditions and can be used as long as the solution it delivers is useful.

Finally, Altendorfer-Kaiser [39] suggest a process model specifically developed for production logistics that is based on iterations as a way to enable progressive building up of analytics capabilities within a supply chain (Fig. 17). They suggest that a change process can be triggered at any stage in a data analytics project provided that the organisation has clearly identified the potential benefits that this would have on the overall project outcome.

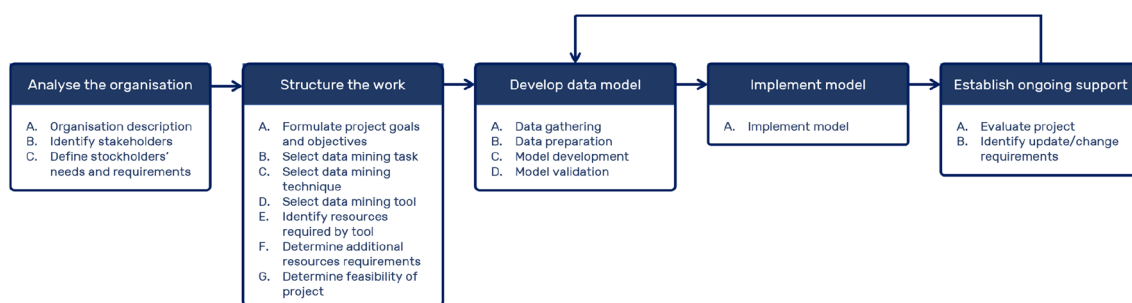


Fig. 14 Data mining process for industrial engineering (DMIE) [37]

Fig. 15 Data science workflow (DMIE) [38]

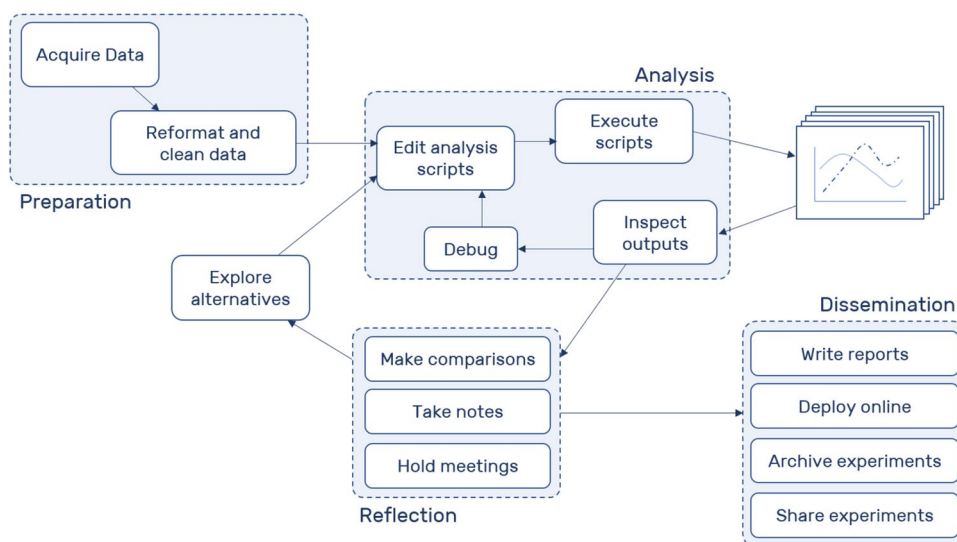


Fig. 16 Foundational methodology for data science [11]

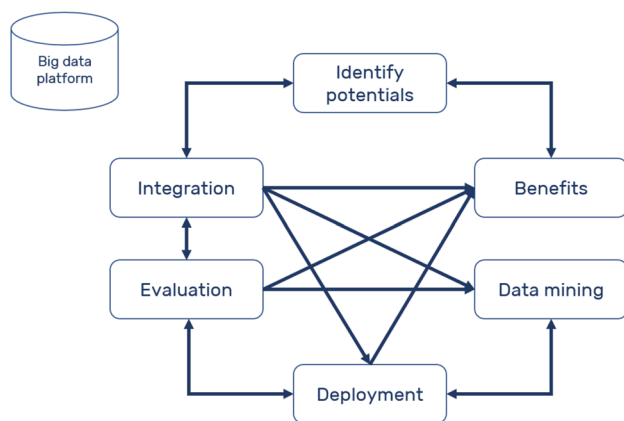
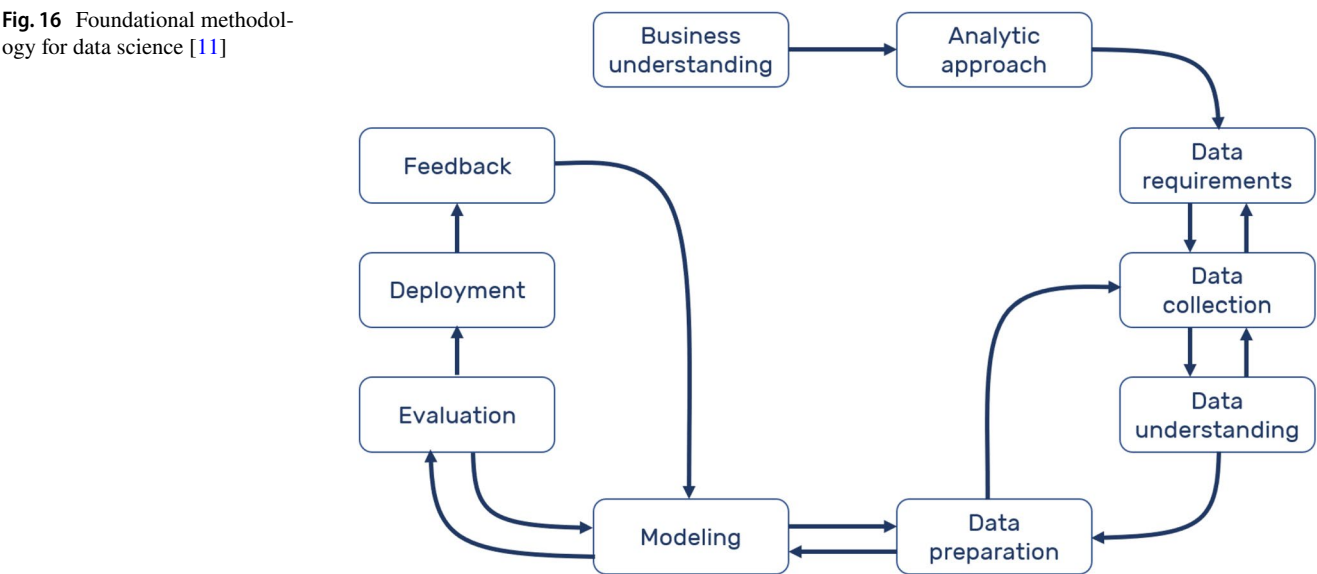


Fig. 17 KD model for production logistics [39]

Interactions in a KD process may refer to both tasks/stages (as highlighted in the SAS analytical lifecycle model, Fig. 6) and people involved. Task interactions and coordination have been recognised as one of the main challenges in a data project [40]; coordination theory and information systems governance should be considered to ensure organisational success of a data science initiative [40]. Interactions between different roles in the process are partially addressed

in models that identify the involvement of various human resources (see “[Relevance of Human Role](#)” section). On the other side, there is a current lack of adoption of mature team process methodologies to conduct data analytics projects; there has been little focus on how a team could effectively work together to do data science, nor much discussion has dealt with the team-based challenges that might occur when a group of people are working on a data science project [15]. RAMSYS is probably the first model that considers challenges deriving from the collaborative nature of a project when people involved are geographically dispersed [34]; the introduction of an information vault (Fig. 18), where the current best understanding of the problem is kept, facilitates sharing of information between the different groups; likewise, the presence of a “model submission” step (which is RAMSYS’ only addition to the CRISP-DM model) makes sure that all groups receive updates regards the currently selected model and modelling progress status.

The theme of interactions (between different actors, i.e. infrastructure experts, data analysts, management, etc.) and iterations (which allow for the integration of new elements and propel continuous improvements) is the focus point of a framework proposed by Schmidt and Sun [26], named Agile-facilitated KD model (Fig. 19). Top layer

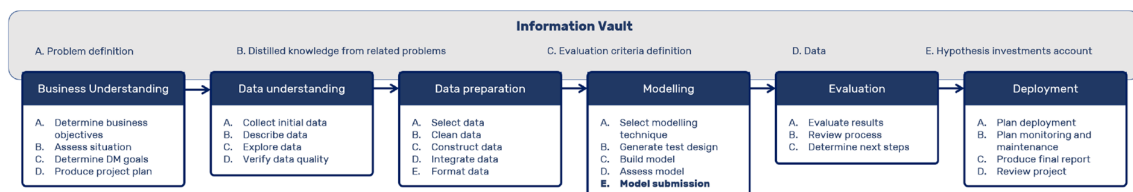


Fig. 18 Rapid collaborative data mining system (RAMSYS) methodology [34]

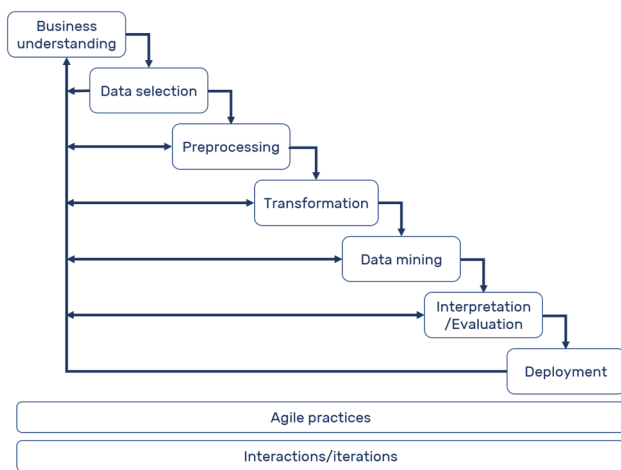


Fig. 19 Agile-facilitated KD [26]

of the framework is the KDD process model with business understanding (which helps data miners to keep focus on the ultimate project goals) and deployment (borrowed from CRISP-DM). They also integrate agile practices in their framework; these are used for conducting activities in each sub-process by including test-first development, daily stand-up meetings, story-telling, pair programming, continuous integration; no specific agile methodology is chosen as appropriate approaches should be selected based on the project type. Examples of implementation of this framework to three projects conducted at an educational institution are presented; details on the choice of agile practices based on the project type and maturity level (i.e. for the third project shell scripts were used) are given. Likewise, details on the activities performed for each phase are discussed with particular emphasis on both interactions and iterations and also implemented approaches borrowed from agile methodologies. The authors find that interactions and iterations especially helped data miners better understand the business requirements; the majority of the iterations were centred around the KDD steps. Although arguably the most important, the deployment step proved to be the most challenging for several factors (i.e. personnel, funding, support to senior managers, etc.); it is also noted that this step is the one most out of control of the data miner as the project ownership is transferred from the IT department to the business units. The authors highlight that automating the steps for data preparation, modelling and evaluation was only possible when they gained confidence in the knowledge discovery process (i.e. through the three projects); this reflects what suggested in classic software engineering approaches.

Saltz et al. [15] compared the performance of four different methodologies/process models by conducting experiments on university students. The metrics used for the comparison were based not only on task output (i.e. project

quality) but also on the team work dynamics, willingness to work together on following projects and satisfaction of individual team members, thus showing the importance of human interactions within a data analytics project.

Data-Focused Process Models

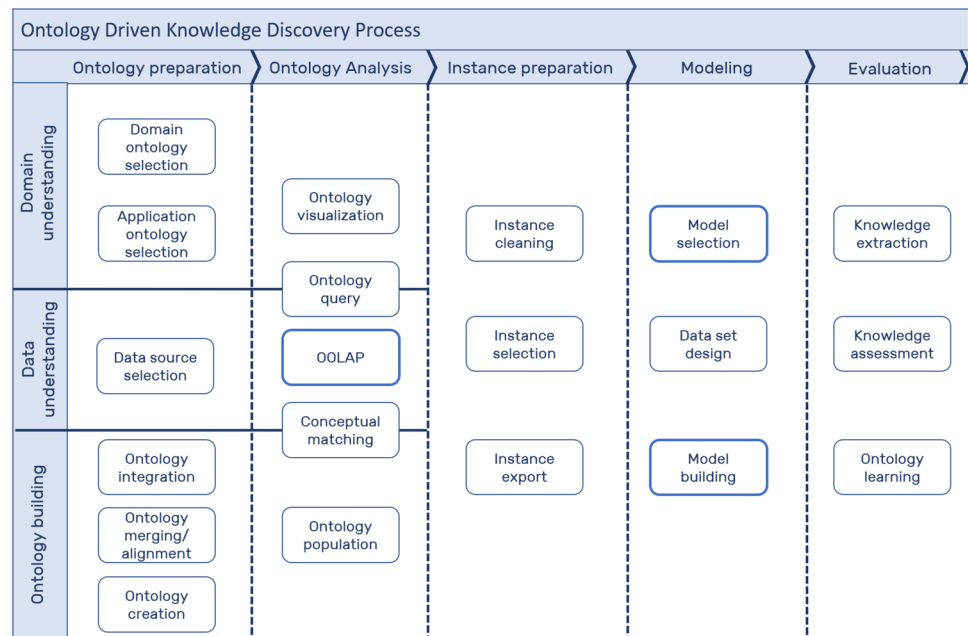
The consideration of various types of data sources, data models and knowledge repositories is also a focal element of KDD process models. Hofmann and Tierney [28] are probably the first one to mention data marts, data warehouses and information and knowledge repositories in their DMLC model (Fig. 10). Data management issues in a KD process are highlighted for all phases of the Data Science Workflow model by Gupta et al. [32] (Fig. 15); for instance, provenance in the preparation phase, output file tracking in the analysis phase, managing notes throughout experiments in the reflection phase and experiment archive in the dissemination phase (i.e. make sure that experiments are replicable and software versions are the same as when the experiments were run) are identified as critical data management elements in all DM projects.

On another level of data source integration, Anand and Büchner [31] adapt the KDD model to web mining projects that use operational and materialised data available on the Internet to gain insights into online customers' behaviour. Knowledge sharing across different domains is the focus of Cios and Kurgan [41] model that is based on technologies like XML, PMML, SOAP, UDDI and OLE DB-DM.

The integration of ontology engineering and KDD is a recent trend in KDD research. It generates a high-level abstraction to the KD process which may allow a better reuse of previous knowledge as well as the creation and evaluation of new knowledge [42]. Domain ontologies can be used to improve the understanding of a problem and support hypothesis-driven analysis and discovery approaches. Ontology-based approaches are also focused on the application of mining techniques in order to automatically or semi-automatically acquire knowledge from data [42]. The Ontology-Driven Knowledge Discovery (ODKD) model developed by Gotttroy [42] borrows CRISP-DM tasks and adapts them to ontology engineering concepts; business understanding and data preparation tasks leverage the semantic description of data (Fig. 20). The use of ontologies also supports the involvement of non-expert users in the data mining process [35].

Recent attention has also been paid to the challenges related to big data analytics. Yew [43] developed a framework that guides people involved in data analytics initiatives to focus on key questions and value creation activities and identify associated goals, assets, tools, methodology. Grady [44] revisits the CRISP-DM model by grouping activities into stages based upon the maturity of the data and

Fig. 20 Ontology-Driven Knowledge Discovery (ODKD) process [42]



the types of processing associated with getting the data to reach that stage of maturity. In her Data Science Edge model (Fig. 21), the project life cycle is typically executed in four distinct phases (i.e. assess, architect, build and improve) and includes five process stages (plan, collect, curate, analyse and act); each stage results in the generation and potential storage and usage of data at increasing levels of maturity; big data aspects are also considered within each stage.

Towards Agile Models

In addition to enhancing existing models, researchers have attempted to integrate knowledge discovery process models into new ones, sometimes borrowing standards from other industries [26].

Marbán et al. [12] observed a parallelism between KD and software engineering (SE); project planning delays, low productivity and failure to meet customers' expectations are characterising the implementation of data analytics

projects. A similar situation was observed in the late 1960s in the software development world and led to the so-called software crisis. In that case, the introduction of new methodologies significantly supported the improvement of software development. Leveraging the experience of SE, Marbán et al. [12] suggested that KD process models should be enriched with tasks and activities that have been integrated with SE development tasks based on empirical evidence. These activities are typical of any engineering process and concern project completeness and quality assurance; they also include organisational, project management and integral processes. The KDD process is framed within more encompassing development processes (Fig. 22). The same authors refine their model and detail the KDD process steps in the so-called Refined Data Mining Process model; this model is conceived to be as complete as possible in terms of process steps and is the result of a critical comparison of KDD process models available in the literature and the selection and combination of key steps from these models

Fig. 21 Data science edge phases, activities and data maturity [44]

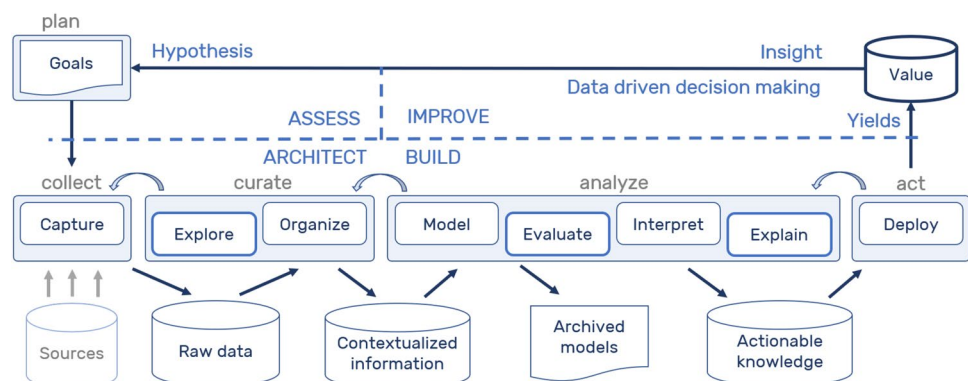
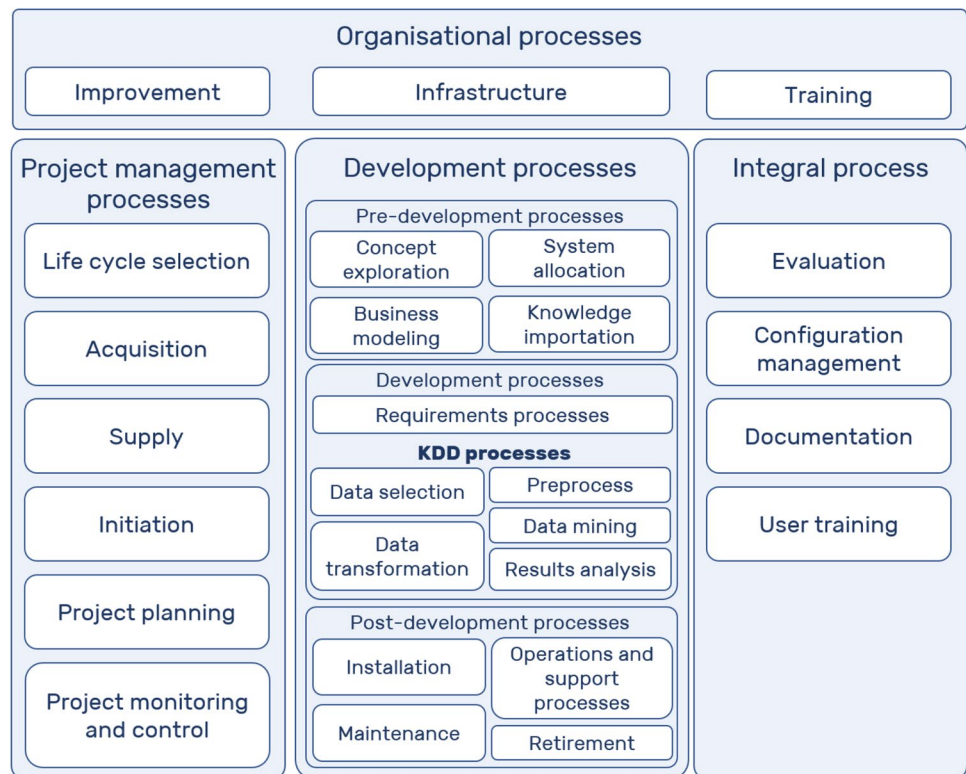


Fig. 22 DM engineering process model [19]

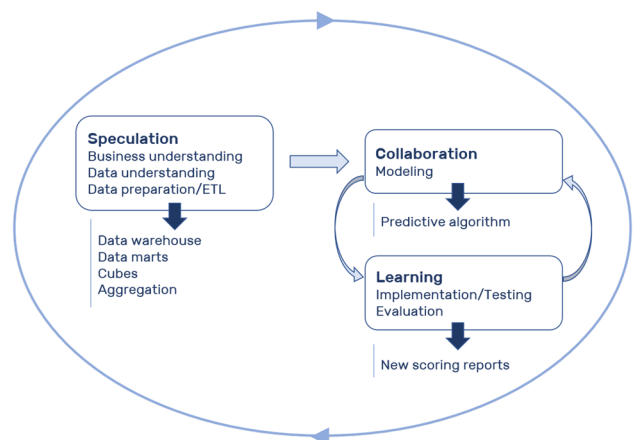
with the intention of avoiding overlaps and covering all tasks included in any data mining and knowledge discovery project development [19].

In the wake of the parallelism with SE, the latest trend in KDD process modelling concerns the integration of agile methodologies. The initial focus in software development concerned the identification of key steps to create a software solution (i.e. classic phased software development model). However, traditional software development models are failing to guide through the delivery of successful projects due to rigid mechanisms with heavy documentation processes which collide with complex, dynamic, ever-changing environments where requirements are uncertain and volatile [45]. SE has evolved towards the use of agile methodologies that acknowledge the importance of group communication and iterations. The thrust to incorporate agile practices in KD processes stems from the consideration that adaptive approaches are best fit when requirements are uncertain or volatile as often happens in data analytics projects due to business's requirements changes, and rapid evolving markets [23]. Beyond their suitability to dynamic contexts, agile methods also value individuals and interactions over processes and tools; they are iterative, lightweight, incremental and emergent and provide support for highly collaborative and cross-functional teams [26]; hence, they are particularly suitable for data analytics initiatives.

The ASD-DM (Adaptive Software Development-Data Mining) process model by Alnoukari and El Sheikh [23]

was one of the first attempts to integrate agile modelling with KDD process modelling. The model focuses on reducing development costs related to customisation or adaptation of mining models to uncertain business objectives. The introduction of a speculation phase recognises the uncertain nature of complex KD problems and encourages exploration and experimentation (Fig. 23). Limitations of this model include the lack of focus on data sources, human resources and deployment phase.

do Nascimento and de Oliveira [45] propose a new KD process model (Agile KDD life cycle) which combines

**Fig. 23** ASD-DM process model [23]

CRISP–DM, KDD and OpenUP (Fig. 24). OpenUP (Open Unified Process) is based on unified process model, that is a use-case driven, architecture-focused software development model. In UP, the architecture provides the framework to guide the system development into iterations, while the use cases define the targets and lead the work of each iteration. OpenUP is a variation of the UP that embraces a pragmatic, agile philosophy that focuses on the collaborative nature of software development. Finally, Schmidt and Sun [26] refer to agile practices as the backbone of their Agile-facilitated KD model (Fig. 19).

Despite the current enthusiasm, empirical evidence on the complexity of the implementation of agile methodologies to KD processes has been given by Saltz et al. [15] that experimented with four different process models:

- *Agile scrum*—based on agile methodology. By-weekly meetings were set to define tasks (sprints)—no change was allowed between two consecutive weeks
- *Agile Kanban*—combines CRISP–DM phases with the pipeline process management approach typical of lean manufacturing Kanban systems;
- *CRISP–DM*;
- *Baseline*—no methodology.

In these experiments, Agile Scrum proved to be the worst methodology, probably due to the complexity of the methodology itself and the lack of a business understanding/planning phase as the definition of sprints led to immediate immersion into coding. CRISP–DM and Agile Kanban

proved to be the two most effective methodologies. The success of Agile Kanban probably relied on the continuous improvement logic that fostered team work. Interestingly, CRISP–DM emerged as a natural way to conduct a data science project; this was further confirmed by the fact that the baseline teams ended up conducting their project in a CRISP-like manner. Nevertheless, the authors highlight that they report an academic experiment conducted on students (not professionals) working part-time on a realistic data project with no customer engagement; motivation and sample size may have affected the results obtained.

Comparative Analysis

A direct comparison of the data mining process models reviewed in the previous sections is reported in Fig. 25. Although not exhaustive and not accurate due to misalignments between process steps, the tables in Fig. 25 highlight similarities between process models and novel contributions of models based on the steps included in their life cycle. The number of process steps varies between three [33] and 17 [19] with an average equal to approximately eight steps. This shows that different approaches exist in terms of chosen level of detail for process models; some authors believe that identifying micro-steps can support better project planning; on the other side, encompassing micro-steps into macro-steps can offer a more practical and flexible view on project phases, while finer details can be provided within the phases [28, 33]. The greater number of steps for the model proposed by Mariscal et al. [19] results from their design approach which is based on a comparative analysis of previous process models; in order to generate a comprehensive process model and provide a detailed task breakdown, the authors combine all relevant steps reported in the models reviewed. The first task in Mariscal et al. [19], namely “lifecycle selection”, is taken from Marbán et al. [12] and suggests that typical waterfall life cycles may not be suitable for all projects and a critical selection of life cycles should be made initially, perhaps taking inspiration from the SE realm. The last step in Mariscal et al. [19] model is taken from Solarte [37] and focuses on establishing ongoing support for maintaining the data mining model and adapting it to evolving business conditions. Although the explicit consideration of ongoing support after the DM model is developed and deployed is distinctive of these two models, the inclusion of a last iteration step is also available in Collier et al. [11, 36, 45]; other models also consider possible final iterations although this is only highlighted through connections between process steps. All other steps in the model by Mariscal et al. [19] detail those included in the CRISP–DM model.

We can identify at least four models that maintain the same process steps as CRISP–DM: Moyle and Jorge [34],

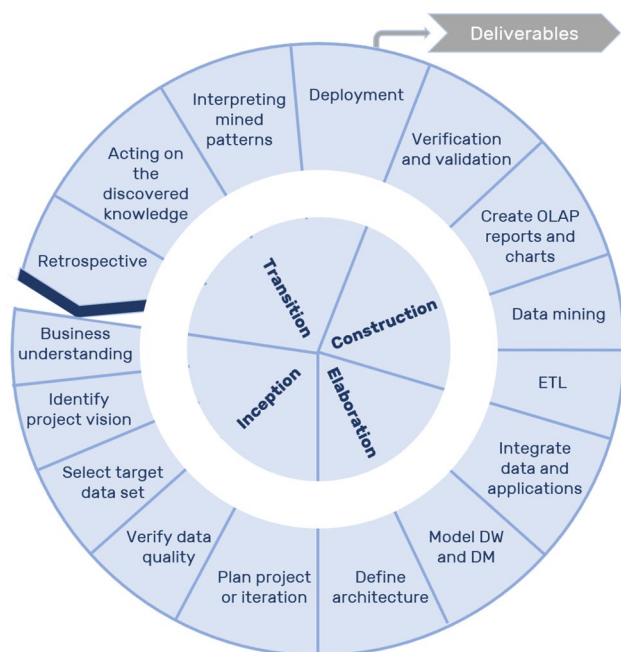


Fig. 24 AgileKDD life cycle [45]

Fig. 25 Comparison of data mining process models based on process steps

Authors	Fayyad 1996	Brachman & Anand 1996	Fayyad detailed 1998	Lee & Kerschberg 1998	Collier 1998	Anand & Buchner 1999	CRISP-DM 2000	Gupta 2000	Cios et al 2000	Moyle 2001	Selarte 2002	Hofmann 2003
Process steps												
		Task discovery	Learning the application domain		Define the objectives	Domain knowledge elicitation	Business understanding	Domain understanding & Data mining requirements	Understanding the problem domain	Business understanding	Analyse the organisation	Business understanding
						Human resource identification					Structure the work	Objectives / hypothesis formulation
						Problem specification						Select/Sample
	Selection	Data discovery	Creating a target data set		Select the relevant business data	Data prospecting	Data understanding	Data selection	Understanding the data	Data understanding		
					Data quality analysis	Methodology identification						
				Planning for learning							Develop data model	
	Pre-processing	Data cleaning	Data cleaning and pre-processing		Clean and transform data	Data pre-processing	Data preparation	Data Cleaning	Data preparation	Data preparation		Pre-processing
	Transformation		Data reduction and projection					Transformation				Transformation
	Data mining	Model development	Choosing the function of DM algorithm	Generate and test hypothesis	Data mining	Pattern discovery	Modelling	Data mining	Build model	Modelling		Data mining
			Data mining									
	Interpretation/evaluation	Data analysis	Interpretation	Determine knowledge relevance	Acquire knowledge	Knowledge post-processing	Evaluation	Knowledge presentation	Evaluation of the discovered knowledge	Evaluation	Implement model	Evaluation
		Output generation		Critique by a panel of experts	Evaluate results			Interpretation				
			Using discovered knowledge	Evolve knowledge data	Deploy results or iterate		Deployment	Application	Using the discovered knowledge	Deployment		Deployment
											Establish on-going support	
Gettgtry 2007	Marban 2007	Alnoukari 2008	Mariscal 2010	Nascimento 2012	Guo 2013	Girardi 2015	Rollins 2015	Rogalewicz 2016	Kormann 2017	Schmidt 2018		
	Life cycle selection		Life cycle selection									
Domain ontology selection	Project Management Processes	Business understanding	Domain knowledge elicitation	Business understanding			Business understanding		Identify potentials	Business understanding		
Application ontology selection			Human resource identification	Identify project vision								
Ontology building	Pre-development processes		Problem specification				Analytic approach		Benefits			
Data source selection		Data understanding	Data prospecting	Select target dataset		Data Modelling	Data Requirements					Data Selection
Ontology analysis			Data cleaning	Verify data quality	Data Acquisition	Data Acquisition	Data collection					
				Plan project								
Instance preparation	Development processes	Data preparation/ETL	Pre-processing	Define architecture	Clean and transform data	Data Validation	Data Understanding					Pre-processing
			Model Data warehouse	Integrate data and application		Data Preparation	Data Preparation	Data preparation				
			ETL									Transformation
Model selection			Data reduction and projection	Choosing the DM task								
Dataset Design		Modelling	Choosing the DM algorithm	Data mining	Data Analysis	Data Analysis	Modelling	Main processing	Data mining	Data mining		
Model building			Build model									
Knowledge Extraction	Integral processes	Evaluation	Evaluation	Create OLAP reports and charts	Reflection	Evaluation	Evaluation		Evaluation	Interpretation/evaluation		
Knowledge Assessment			Interpretation	Verification and validation	Explore Alternatives							
		Implementation	Deployment	Deployment	Deployment			Post-processing	Deployment	Deployment		
				Interpreting mined patterns								
				Acting on discovered knowledge								
Ontology Learning	Post-development processes		Automate	Retrospective			Feedback					
			Establish on-going support									

Cios and Kurgan [41], Alnoukari and El Sheikh [23] and Schmidt and Sun [26]. Moyle and Jorge [34] adapt CRISP-DM to geographically distributed projects and add a model submission sub-step in the modelling phase to ensure all project participants are kept updated on the project progress. This is achieved via an information vault, which consists of a data, knowledge and good practice repository shared among project stakeholders. Cios and Kurgan [41] use slightly different names for their steps and introduce different iterations between tasks, and they explicitly mention knowledge as the result of the data mining process. Alnoukari and El Sheikh [23] and Schmidt and Sun [26] revisit the CRISP-DM model from an agile perspective; Alnoukari and El Sheikh [23] abstract the first three steps of CRISP-DM into a speculation phase which indicates that almost a tentative approach should be taken in formulating

the business problem, identifying the data requirements and building a data mining model; this is because of the volatile nature of business objectives in highly dynamic business environment. With respect to CRISP-DM, Schmidt and Sun [26] separate the data preparation step into pre-processing and transformation, and they keep all other steps the same as CRISP-DM, but they add multiple iterations and interactions between steps and introduce an inherent agile concept to make the CRISP-DM model more adaptable to different business conditions.

Next, we will consider each single step in the CRISP-DM model and analyse how they translate into the other models reviewed. The business understanding step is available in most models although at times it is referred to using different names (i.e. “learning the application domain” in Fayyad et al. [21], “understanding the problem domain” in Cios and

Kurgan [41]). This step is missing in Lee and Kerschberg [30], Guo [38] and Girardi et al. [35]. In Lee and Kerschberg [30] and Girardi et al. [35], this choice is justified by the integration of domain experts in the data mining process; hence, an initial business understanding phase is not explicitly formalised but is ensured through the committed involvement of subject matter experts that will be able to transfer their knowledge and validate the relevance of discovered patterns through ongoing feedback. On the other side, Guo [38] adopts the perspective of scientists, engineers and researchers who need to program for their jobs; hence, they already possess domain knowledge and have clear objectives to pursue and hypotheses to validate through data mining. Before CRISP-DM was introduced, Anand and Büchner [31] had identified three tasks involved in business understanding: “domain knowledge elicitation”, “human resource identification” and “problem specification”, which are also reported in Mariscal et al. [19]. Objectives definition and problem specification are generally the two main outcomes of the business understanding step, which precede the data understanding phase. However, Hofmann and Tierney [28] introduce the objectives and hypotheses formulation after the data understanding steps; this modification can be valid when business objectives are not clear and preliminary data analyses can help formulate business problems more clearly. Alternatively, it can apply to situations whereby constraints on data availability should be taken into account while defining data mining objectives (i.e. investments in data infrastructure are not permitted). The model proposed by Gottgroy [42] does not directly compare with other process models as it presents a matrix structure that combines the typical CRISP-DM steps with tasks related to ontology building and maintenance; the focus is on structuring knowledge and data related to a data mining project in a way that can be reusable.

The data understanding step is often further detailed in the models reviewed and referred to using different names. It tends to include data acquisition, which is functional to data quality analysis. Data discovery [29], data prospecting [19, 31], data requirements [11], data modelling [35], select target data set/ sample data or data acquisition [11, 26, 28, 32, 36, 38, 42, 45], data validation/data quality analysis [35, 36, 45] can all be considered part of the data understanding phase. Data preparation as a generic step comprising all tasks required to make the available data ready to be analysed is reported in many models [11, 23, 27, 33–35, 41, 42]. It is also referred to as data pre-processing [19, 26, 28, 31], ETL [23, 45], data cleaning [12, 29, 32] and data transformation [26, 28, 32]. In this process phase, do Nascimento and de Oliveira [45] also introduce steps related to data architecture definition and integration.

The modelling step is common to all process models. Data analysis, data mining, model building, pattern

discovery and main processing are all terms used to refer to this step. Model selection [42] or choosing the DM task and DM algorithm [19, 21] can precede the actual model development and is highlighted separately in some process models. In this regard, the use of the word “knowledge” as the purpose and result of a data mining process is interesting but only explicitly mentioned in a few models.

The evaluation step is also included in all models. Beside evaluation, typical denominations for this step include interpretation, determine knowledge relevance, reflection, knowledge assessment, verification and validation. Interestingly, Brachman and Anand [29], Gupta et al. [32] and do Nascimento and de Oliveira [45] introduce a knowledge presentation step to highlight the importance of conveying the data mining output in a form that is understandable to non-technical people. In particular, do Nascimento and de Oliveira [45] explicitly mention that OLAP reports and graphical charts should be generated to democratise the mining insights, which is very much in line with the principles and popularity of visual analytics.

Finally, the deployment step is generally included in process models but is missing in Anand and Büchner [31], Gottgroy [42] and Girardi et al. [35].

Discussion and Conclusions

The biggest barriers to the adoption of analytics are organisational [6]. Large technology investments are used by companies as the sole strategy to introduce analytics in their organisation; however, cultural changes and a clear strategic vision are key elements to pave the road to analytics success. Indeed, leadership sceptical of impact and uncertainty around the steps to take to derive business insights from data represent some of the managerial impediments to analytics adoption [6]. In this regard, process models can provide fundamental support to guide the implementation of analytics initiatives. Effective process models should be used by business managers and data analysts to ensure that fundamental aspects of analytics initiatives are not neglected so that project management-related failures can be minimised. Classic process models are illustrated in “[Classic KDD Process Models](#)” section. Despite having being on the scene for two decades, the CRISP-DM model remains the *de facto* industry standard. The main merit of this model is to introduce a business understanding phase, which leads to define clear business objectives, and a data understanding phase, which highlights current data availability and defines data requirements. The main limitations of the CRISP-DM model include: lack of focus on the human role; adoption of a waterfall life cycle (i.e. limited iterations or interactions between phases); lack of consideration of various data

sources; and lack of integration of software development processes. Based on these limitations, research streams for KD process models' development have emerged.

- *The evolution of the human role*—beside classic data-centric models, human-centric models that clearly define the role of different players involved in the various phases of a project have been developed (see “[Relevance of Human Role](#)” section). A clear identification of human resource requirements and their skills represents a key step to avoid project mismanagement. For instance, recent research has highlighted that beside the obvious need for people with data science skills, organisations should also consider complementary roles, such as analytics translators; these combine data science knowledge with industry or functional expertise and are able to translate business questions into modelling specifications and, vice versa, analytics results into business insights [6]. More interestingly, a trend in process models' development that is also common to all analytics aspects has emerged and concerns the involvement of non-IT expert as key players in data analytics projects. Means to make analytics more amenable to a non-technical audience should be catered for while planning a data analytics project. For instance, knowledge structuring methods, such as ontologies, and data analytics platforms with seamless integration in IT systems should be considered as potential means to minimise the IT experts involvement in a project and maximise the domain experts' contribution, especially in contexts where domain expertise cannot be captured by data. In this regard, the citizen data scientist concept is also becoming popular [46].
- *The definition of iterations and interactions*—these concern connections between phases and actors within a KD project life cycle (see “[Iterations and Interactions](#)” section). The reason why iterations and interactions have surged to become critical aspects of process models relies on the necessity of making analytics projects execution as dynamic as possible. Iterations serve the purpose of both skipping project phases without the need to follow process steps in a consecutive fashion and return to previous steps without waiting for the project completion if modifications to the project requirements have occurred in the interim. This is especially useful in very dynamic environments where business needs are volatile. The need for introducing iterations also aligns with the continuous improvement logic that triggers project cycle repetitions to maintain the analytics artefacts updated and prolong their usability in time. The focus on interactions refers to the relevance of various human roles within a project and highlights that effective communication is a key element to project success, especially for large size projects with geographically dispersed interest groups.
- *The relevance of data and knowledge repositories*—this element encompasses different aspects related to data in a project life cycle (see “[Data-focused process models](#)”):
 - Data management—i.e. provenance, security, governance, etc.;
 - The inclusion of various data sources—beside databases and data lakes, web data currently represents a fundamental source of information; for instance, social media aware manufacturing is a latest trend that aims at making product manufacturing a collaborative and collective process [47];
 - The use of ontologies—to structure knowledge elicited from data, make it reusable and automatise the data mining process;
 - Big data challenges.
- *The integration of software engineering / agile methodologies*—(see “[Towards Agile Models](#)” section). This is in response to the increasing dimension of data analytics projects which are becoming software development projects at all effects. A parallel with software development history has also been highlighted in the literature, and the opportunity to transfer lessons learnt from software engineering to data analytics has been considered and realised. When a software development perspective is used, the KDD process becomes a mere step in a more comprehensive software development framework that encompasses project management and development processes. In this context, inspiration has been taken from agile methodologies that perfectly suit highly dynamic environment; indeed, volatile objectives which are typical of exploratory analytics projects make classical software development approaches too rigid and cumbersome to follow.

The review conducted shows that although research efforts have been conveyed into overcoming some intrinsic limitations of classic process models, improvements have been marginal in the sense they have focused on specific aspects (i.e. human role, ontologies, agile methodologies, etc.). In our opinion, a more holistic approach to process model development should be taken and the various research directions identified in this review should converge into one to offer better support to analytics project management. In order to minimise the risk of analytics project failures, process models should also be enriched with methodology aspects that could offer guidance on how to develop the process steps, clarify the inherent requirements and provide means to support the prediction of the potential impact of analytics initiatives.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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